National Park Service Analysis

Biodiversity in National Parks

Conservation Status & Species Observation Analysis



Focus: Cross-park species vulnerability & observation frequency analysis

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Data Overview (species_info.csv & observations.csv)

Dataset Composition

- Evaluated 5,824 species records across 7 taxonomic categories.
- 191 species hold active federal conservation listings, while 5,633 records required imputation as No Intervention.

Status Hierarchy

- The vast majority of protected species fall under Species of Concern rather than active Endangered status.
- Observations cover a 7-day rolling window across 4 national parks totaling tens of thousands of sightings.

Data Cleaning Actions

- Imputing missing conservation_status values to prevent omission of unlisted, healthy species.
- Created binary indicator is_protected to evaluate protected vs. non-protected proportions across categories.

Conservation Vulnerability by Category

17.05%

Mammals

- Highest proportion of species under conservation protection.
- Heavily weighted toward Species of Concern and active Endangered designations.

15.37%

Birds

- Second highest protection rate across all taxonomic groups.
- High species richness with substantial presence across all four park habitats.

6.41%

Reptiles

- Lower proportion under federal management showing greater baseline population stability.

~1.08%

Plants

- Vascular plants constitute the bulk of total records (~4,470 species) but maintain the lowest rate of intervention.

Hypothesis Testing: Chi-Square (χ^2) Analysis

Methodology

Conducted contingency tests to assess whether protection differences between categories are statistically significant.

Test Parameters

Standard significance threshold set at $\alpha = 0.05$

Test Outcomes

Mammals vs. Birds

$p \approx 0.6876$

FAIL TO REJECT H_0

Difference is not statistically significant

Mammals vs. Reptiles

$p \approx 0.0384$

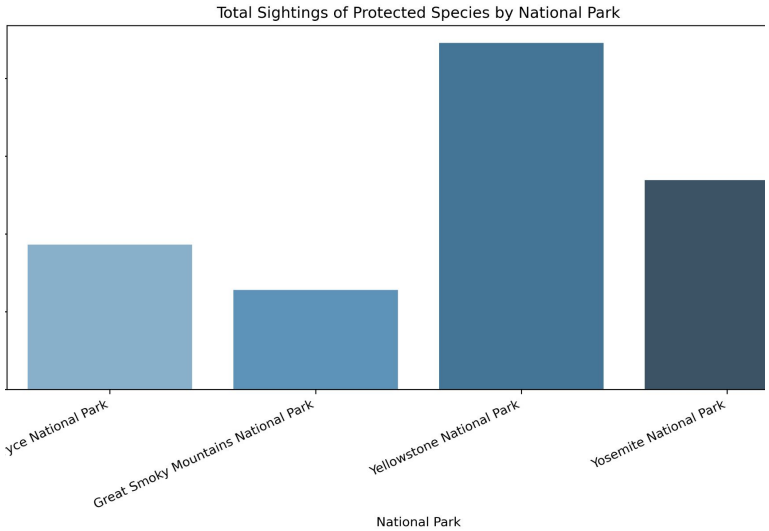
REJECT H_0 ($\alpha = 0.05$)

Difference is statistically significant

Scientific Implications

- **Shared Focus:** Mammals and birds face comparable systemic ecological threats, justifying shared broad-spectrum conservation focus.
- **Conservation Urgency:** The significant gap between mammals and reptiles indicates vertebrate endotherms face elevated conservation urgency.

Park Sightings Breakdown



44,583

Yellowstone

Leads in protected sightings due to extensive area and active monitoring.

26,989

Yosemite

Record protected species sightings.

18,602

Bryce Canyon

Recorded protected species sightings.

12,796

Great Smoky Mountains

Record protected species sightings.

Recommendations & Next Steps Determination

Targeted Interventions:

- **Prioritize Mammal & Bird Habitats:** Channel active ranger monitoring and habitat protections toward mammalian and avian migration corridors.
- **Standardize Density Metrics:** Transition from raw observation counts to sightings per square mile to normalize for park acreage disparities.

Foot & Mouth Disease Sample Size

- **Baseline Assessment:** Evaluated sample size requirements for detecting a meaningful shift in disease prevalence among ungulate populations.
- **Statistical Rigor:** Applied power analysis (80% power, α = 0.05, expected minimum detectable effect) to ensure future park epidemiological surveys collect sufficient animal observations to detect transmission spikes.